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A study of plants used by ethnic tribal population of Ranchi District against Malaria

Swet Nisha Sangeeta[a]* & Ashok Kumar Nag[a]
[a]*Department of Botany, Ranchi College, Ranchi, 834001, Jharkhand India

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Abstract : Malaria is one of the most harmful diseases in Ranchi district.It is known that Malaria is a mosquito borne infectious disease of humans. It is caused by the genus *Plasmodium*. In Ranchi district there are many plants which are being used by the ethnic tribal population to cure malaria. In the present paper, 38 plant species have been identified with the help of herbal practitioners which are claimed to be effective against malaria. There are 32 castes which are enlisted in the category of the scheduled tribes in jharkhand.Out of these 32 castes are the original inhabitats of the present Ranchi District who have been using some plants traditionally to treat malaria.

Keywords: Malaria, Tribes, *Plasmodium*.

INTRODUCTION

Ranchi district is one of the twenty four districts of Jharkhand state of India. Ranchi city is the capital of Jharkhand state . It was established as a district in 1899 when it was a very big district have been carved out of this district over the years.Presently it has only two sub divisions namely Ranchi and Bundu.Ranchi district is further divided into 11 blocks ; Angara, Burmu, Bero, Chanho, Kanke,Lapung, Mandar, Namkum, Ormanjhi, Ratu and silli. Bundu sub-division comprises 3 blocks : Bundu, Sonahatu and Tamar. There are 2145 no. of villages in Ranchi District. There are as many as 32 different tribes in Jharkhand state of India. Main tribes of these area are Munda, Oraon, Santhal, Ho, Kharoa, Sabbar, Birhor, Karmali, Chero. Malaria is a Mosquito-borne infectious [2] disease of humans and other animals caused by parasitic protozoans of the genus *Plasmodium*. Commonly, the disease is transmitted via a bite from an infected female *Anopheles* mosquito, which introduces the organisms from its saliva into a person's circulatory system. Five species of *Plasmodium* can infect and be transmitted by humans. The vast majority of deaths are caused by *P. falciparum* & *P.vivax* . While *P. ovale*, and *P. malariae* causes a generally milder form of Malaria that is rarely fatal. It includes fever and flu-like illness, including shaking chills, headache, muscle aches, tiredness. Nausea, vomiting and diarrhoea may also occur. For example: In Angarha of Ranchi district where 61 uncomplicated cases of *P. Falciparum* malaria were registered out of these early treatment failure was seen in 3 cases and late treatment failure in 14 cases.

MATERIALAND METHOD

Firstly field trip were conducted and local Vaidyas were contacted and people suffering from malaria and the herbal practitioners were personally contacted for gathering information by a proper questionnaire drafted for a specific purpose. The first hand information regarding the plants and the methods of administration were identified and the plants which are said to cure malaria were enlisted.This plants [5] were verified by the many authors. [1,3,4,6,7,8,9,10]

*Corresponding author :
Phone: 08252989904
E-mail : Sangeeta.swetnisha@gmail.com

Table 1: Details of plants used used by ethnic tribal population.

S. No.	Botanical Name	Family	Common Name	Plant Parts Used	Method of Using
1	*Andrographis paniculata L.*	Acanthaceae	Kalmegh	Bark and leaves	taken orally with water
2	*Caesalpinia crista L.*	Caesalpiniaceae	Latakaranja	Stem, bark	boil with water
3	*Allium sativum L.*	Amaryllidaceae	Garlic	Bulb	tincture, decoction
4	*Carica papaya L.*	Caricaceae	Papaya	Fruit, leaves	infusion
5	*Azadirachta indica A.juss*	Meliaceae	Neem	Bark, leaves	decoction
6	*Artemisia annua L.*	Asteraceae	Sweet worm wood	Stem	aqueous extract
7	*Ocimum sanctum L.*	Lamiaceae	Tulsi	Leaf	plant extract
8	*Cymbogon citrates (DC.)stapf*	Poaceae	Lemon grass	Leaf	vapours of decoction
9	*Coriandrum sativum L.*	Apiaceae	Dhania	Seeds	decoction
10	*Tinospora cordifolia L.*	Menispermaceae	Guduchi	Stem	aqueous extract
11	*Cinnamon cinchonaL.*	Rubiacease	Quinine	Stem	aqueous extract
12	*Tamarindus indicaL.*	Fabaceae	Imli	Fruit	infusion
13	*Psidium guajavaL.*	Myrtaceae	Guava	Fresh or dried leaves, bark, fruit	boil & decoction
14	*Allium cepa L.*	Amaryllidaceae	Onion	Bulb, Leaves	tincture, decoction
15	*Hibiscus cannabinus L.*	Malvaceae	Kudrum	Fruit, leaves	decoction
16	*Musa sinensis L.*	Musaceae	Banana	Fruit	infusion
17	*Centella asiatica (L.) urban*	Mackinlayaceae	Beng Sag	Leaves	decoction
18	*Plumbago zeylanica L.*	Plumbaginaceae	Chitrak	Roots	by grinding & make Juice
19	*Mucuna pruriens(L)DC*	Fabaceae	Alkusi	Fruit	infusion
20	*Aegle marmelos (L.)*	Rutaceae	Bel	Fruit	infusion
21	*Cannabis sativa L.*	Cannabaceae	Bhang	Leaves	decoction
22	*Achyranthes aspera L.*	Amaranthaceae	Chirchiri	Whole plant	boiling, decoction
23	*Ageratum conyzoides L.*	Asteraceae	Bhunsri ghas	Leaves	decoction
24	*Alstonia scholaris L.R.Br.*	Apocynaceae	Chitwan	Bark, Leaves	decoction
25	*Asparagus racemosus Wiild.*	Asparagaceae	Satawar	Stem	boil
26	*Phyllanthus fraternus L.*	Phyllanthaceae	Bhuin anwla	Leaves	decoction
27	*piper longum L.*	Piperaceae	Pipli	Leaves	decoction
28	*Sphenocentrum jollyanumL.*	Menispermaceae	Akerejupon	Roots	juice
29	*Rauvolfia vomitoria Afzel.1817*	Apocynaceae	Asofeyeje	Roots, barks, leaves	infusion
30	*Mangifera indica L.*	Anacardiaceae	Mango	Bark, leaves	decoction
31	*Tithonia diversifolia (Hemsl.)*	Compositae	Tree Marigold	Leaves, stem twigs	decoction
32	*Zingiber officinale Roscoe*	Zingiberaceae	Ginger	Rhizome	decoction, tincture
33	*Cucuma longa L.*	Zingiberaceae	Turmeric	Rhizome	tincture,decoction
34	*Citus aurantifolia L.*	Rutaceae	Lime	Root, bark, stem twigs, leaves, fruit	decoction,boil
35	*Citrus paradisi L.*	Rutaceae	Grape	Leaves, root	tincture
36	*Gossypium barabadenseL.*	Malvaceae	Cotton	Leaves	decoction
37	*Anacardium occidentale L.*	Anacardaceae	Kaju	Bark, leaves	decoction
38	*Solanum nigrum L.*	Solanaceae	Black night shade	Leaves	decoction

RESULT AND DISCUSSION

The survey done on the tribal population with a view to obtain knowlege regarding the plants or plant parts, the ethnic population is using about 38 different species of plants for getting relief from Malaria. The plant parts are used in various ways in the form of juice, paste, powder, decoction or extract.

REFERENCES

1. **Saxena, S: Pant, Jain, D.C. and Bhakuni, R.S. 2003**. Antimalarial agents from plant:sources, Curr, sci, 85 (9) : 1314-1329.
2. **Cox, F, E. 2010.** History of the discovery of the malarial parasites and their vectors. Parasites and vectors 3(I) : 5.
3. **Willicox, M.L. and Bodekar, G. 2004.** Traditional herbal medicines for malaria British Med J. 329 (7475) : 1156-1159.
4. **Bhattarai, N.K. 1997**. Folk Herbal Medicines for some Major Diseases in Nepal and their Therapeutic Efficacy in Relation to the Ayurvedic and Modern Systems. D. Sc Thesis, Ranchi University, Ranchi.
5. **Haines, H.H. 1921-25**. The Botany of Bihar and Orissa. part I-VI. West Newman, Ltd. London.
6. **Hembrom, P.P. 1993.** Adivasi-Ausadhi (Horopathy), vol I-VII. Paharia Sewa Samiti, Pakur, Jharkhand.
7. **Iyer, S.R. 1995**. Indian Medicinal Plants. Orient Longman Ltd. Madras.
8. **Balkrishna, Acharya. 2008**. 'Ayurved': Jari Buti Rahasya,: Divya Prakashan, Patanjali Yogpeeth, Haridwar.
9. **Hembrom, L.S. 2003.** Jari Buti Chikitsa Padhati,: Tribal Tradition, Satia, Pakur, Jharkhand.
10. **Chopra, R.N., Nayar, S.L. and Chopra, I.C. 1956**. Glossary of Indian Medicinal Plants .C.S.I.R. New Delhi

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